

Shifan Yu

Xi'an Jiaotong University, Xi'an, Shaanxi, China

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RESEARCH INTERESTS

Quantum Information, Cosmology, Gravitational-Wave Physics, Computational Physics

EDUCATION

Xi'an Jiaotong University, Xi'an, China

Sept. 2024 – June. 2028 (Expected)

Bachelor of Science in Applied Physics

Cumulative GPA: 3.43/4.30

*Elite Programs: Selected Member of **Yan Jici Talented Class** (Co-built with Institute of Physics, CAS)*

*& **Wu Youxun Talented Class** (Co-built with China Institute of Atomic Energy)*

ACADEMIC EXPERIENCE

Quantum Computing and Quantum Error Correction Research

Xi'an, China

Research Intern (Superv. by Prof. Changchun Zhong)

Jan. 2025 – Jul. 2025

- Conducted a 6-month survey of major quantum computing paradigms, benchmarking theoretical foundations, architectures, and open challenges.
- Investigated Quantum Error Correction, analyzing noise models, quantum encoding, syndrome measurement, and recovery strategies for fault tolerance.
- Authored a 10,000-word Chinese review synthesizing 76 academic sources, bridging physical-layer quantum noise and logical-qubit protection.

Cold Atom Laboratory

Xi'an, China

Research Intern (Superv. by Prof. Tao Chen)

Oct. 2025 – May. 2026

- Dedicated 20 hours per week to independently constructing a 780 nm external-cavity diode laser, executing beam collimation, grating optical feedback, polarization control, and power distribution.
- Optimized external-cavity optical feedback via iterative grating adjustments and beam alignment, suppressing multi-mode competition to ensure stable, single-frequency operation.
- Characterized laser performance, determining a 143 mA threshold and delivering 40 mW output power to drive downstream frequency-stabilization and cold-atom experiments.

PROJECTS

Vacuum-Gauge Layout Optimization in Ultra-High-Vacuum Systems

Xi'an, China

Core Member (Superv. by Assoc. Prof. Jie Wang)

Jan. 2026 – June. 2026

- Engineered a test-particle Monte Carlo (TPMC) solver to simulate molecular transport and diffuse wall reflections, accurately resolving vacuum conductance across complex UHV geometries.
- Validated simulation accuracy against analytical benchmarks at 10^5 particles/surface, constraining relative errors to 1.57% for transmission probability and 1.70% for conductance.
- Benchmarked 9 tube geometries and formulated a mesh-sizing criterion based on aspect ratio, accelerating simulations for vacuum-gauge placement optimization.

Computational Holographic Reconstruction on Non-Parallel Planes

Xi'an, China

Project Leader (Superv. by Prof. Rui Feng Liu)

May. 2026 – Present

- Implemented an angular-spectrum rearrangement framework for non-parallel diffraction, constructing an end-to-end pipeline for phase-only hologram optimization and wavefield propagation.

- Built and calibrated a 632.8 nm He–Ne/SLM/CCD optical testbed, achieving high-fidelity holographic projection on a 35° tilted plane with a target-image correlation coefficient of 0.770.
- Extended the architecture to multi-plane dynamic holography, multiplexing up to 4 targets across distinct depths and orientations into a single phase hologram with Dice scores up to 0.851.

Thermo-Optical Design of a Four-Mirror fabry–pérot Cavity for HEPS

Student Researcher (Superv. by Dr. Haodi Wang)

Remote

May. 2026 – Present

- Designed a 1064-nm non-planar 4-mirror enhancement cavity for laser–Compton scattering at HEPS, modeling its 3D geometry to support 200 kW circulating power and a $15.6 \times 4.8 \mu\text{m}$ interaction spot.
- Engineered a self-consistent thermo-optical cavity model, validating simulation fidelity against 6 published benchmark suites with key deviations constrained within 6.53%.
- Resolved a thermally stable operating regime at 198.6 kW (retaining 99.3% target power), constraining all mirror temperature rises under 1 K while matching target beam size within 0.76%.

Quantum Search for Space-Based Gravitational Waves

Student Researcher (Superv. by Prof. Jibo He)

Remote

June. 2026 – Present

- Reproduced a quantum matched-filtering search over a 131,072-template bank and audited the quantum-counting and template-retrieval stages underlying four published simulation figures.
- Traced the anomalous long-tailed query costs to an incorrect conversion; derived and implemented the correct mapping and conducted 1,000-trial Monte Carlo sweeps.
- Benchmarked two template-retrieval strategies to regenerate 4 corrected figures, demonstrating tightly bounded oracle-query costs with substantially suppressed tail latency.

PUBLICATIONS & INTELLECTUAL PROPERTY

Journal Papers

Submitted

- **Yu, S.** (2026). Detector–temperature gauge and robust calibration in collisional quantum thermometry. *Submitted to Physics Letters A*.

Reviewed

- **Yu, S.** (2026). The Macroscopic Heisenberg Microscope: The Standard Quantum Limit in Gravitational-Wave Detectors. *Reviewed at American Journal of Physics*.

Published

- **Yu, S.** (2026). Probabilistic controlled quantum teleportation of an arbitrary qubit via a four-qubit asymmetric entangled state and its Qiskit-based Monte Carlo simulation. *International Journal of Theoretical Physics*, **65**, 236. doi:10.1007/s10773-026-06433-y.
- **Yu, S.**, Guo, F., & He, J. (2026). Erratum: Quantum search for gravitational wave of massive black hole binaries [Phys. Rev. D 112, 083004 (2025)]. *Physical Review D*. doi:10.1103/x3pz-ng9z.

Computer Software Copyrights

- **Yu, S.** (2026). *Optical Interference Phenomenon Observation and Analysis Platform V1.0*. Cert. No.: 17634738, Reg. No.: 2026SR0420457.
- **Yu, S.** (2026). *Atomic Physics Experiment Data Recording and Analysis Platform V1.0*. Cert. No.: 17333408, Reg. No.: 2026SR0119127.
- **Yu, S.** (2026). *Optical Experiment Light Path Design and Data Management System V1.0*. Cert. No.: 17333437, Reg. No.: 2026SR0119156.

HONORS & AWARDS

Meritorious Winner , Mathematical Contest in Modeling (MCM)	2025
Third Prize , XJTU Mathematical Modeling Competition	2025
Third-Class University Scholarship , Xi'an Jiaotong University	2025
Honorable Mention , 37th Tengfei Cup Innovation & Entrepreneurship Competition	2026
Team First Prize & Individual 4th Place , Yan Jici Class Academic Competition, XJTU	2026
Honorable Mention , 11th Shuwei Cup University Mathematical Modeling Challenge	2026
Team Second Prize , Yan Jici Elite Class Academic Competition, Institute of Physics, CAS	2026
First Prize , 4th Northwest Regional College Student Physics Experiment Competition	2026

OTHER EXPERIENCES

International Summer School on Mathematics and Physics	Changchun, China
<i>Participant (Jilin University & Northeast Normal University)</i>	Aug. 2025
The 2nd AI4Science Beijing Meetup 2026	Beijing, China
<i>Participant (YMSC, Tsinghua University)</i>	Jun. 2026
International Summer School on Quantum Materials and Information	Taiyuan, China
<i>Participant (Shanxi University)</i>	Jul. 2026
Summer School on Mathematical Principles of Quantum Science	Shanghai, China
<i>Participant (East China Normal University)</i>	Aug. 2026
Fudan Particle Physics Summer School 2026	Shanghai, China
<i>Participant (Department of Physics, Fudan University)</i>	Aug. 2026
The 8th International Symposium on Quantum Physics and Quantum Information Science (QPQIS 2026)	Beijing, China
<i>Participant (Beijing Academy of Quantum Information Sciences)</i>	Oct. 2026

SKILLS

- **Programming Languages:** Python, C, MATLAB.